

- Discuss the functional features of Battery Management Systems in AV/EV battery module with relevant schematic diagram.
- Compare and contrast the differences in power output, charging speed, and connector types between Level 1, Level 2, and DC Fast Charging (Level 3) EV charger standards. How do these differences impact the suitability of each charger type for different environments?
- Explain how the CAN (Controller Area Network) protocol is applied in modern vehicles to manage communication between the engine control unit (ECU) and other subsystems. Provide examples of specific use cases where CAN protocol enhances vehicle performance and safety.
- Demonstrate how an OBD-II diagnostic tool can be used to monitor real-time vehicle data, such as fuel efficiency, engine RPM, and coolant temperature.